

5(a)(i)	conversion (from a.c.) to d.c.	B1
5(a)(ii)	full-wave (rectification)	B1
5(b)(i)	P labelled – and Q labelled +	B1
5(b)(ii)	V_{OUT} scale labelled 4 and 8 on the 2 cm tick marks	B1
	$T = 2\pi / \omega$ $= 2\pi / 25\pi$ $= 0.08 \text{ s}$	C1
	t scale labelled 0.02, 0.04, 0.06, 0.08, 0.10, 0.12 on the 2 cm tick marks	A1
5(c)(i)	correct symbol used for capacitor and capacitor connected in parallel with the $1.2 \text{ k}\Omega$ resistor.	B1
5(c)(ii)	straight lines or curves, with negative decreasing gradients, drawn between adjacent peaks, from top of first peak to meet line going up to next peak	B1
	lines, from one peak to the line going up to the next peak, show a drop in p.d. of $1\frac{1}{2}$ small squares	B1
5(c)(iii)	$V = 0.90 \times 6.0 (= 5.4 \text{ V})$ or discharge time (for each cycle) = 0.034 s	C1
	$V = V_0 \exp(-t/RC)$ $5.4 = 6.0 \exp[-0.034 / (1.2 \times 10^3 \times C)]$	C1
	$C = 2.7 \times 10^{-4} \text{ F}$	A1